

Assignment 2 Writeup: A Little Slice of pi

Introduction:

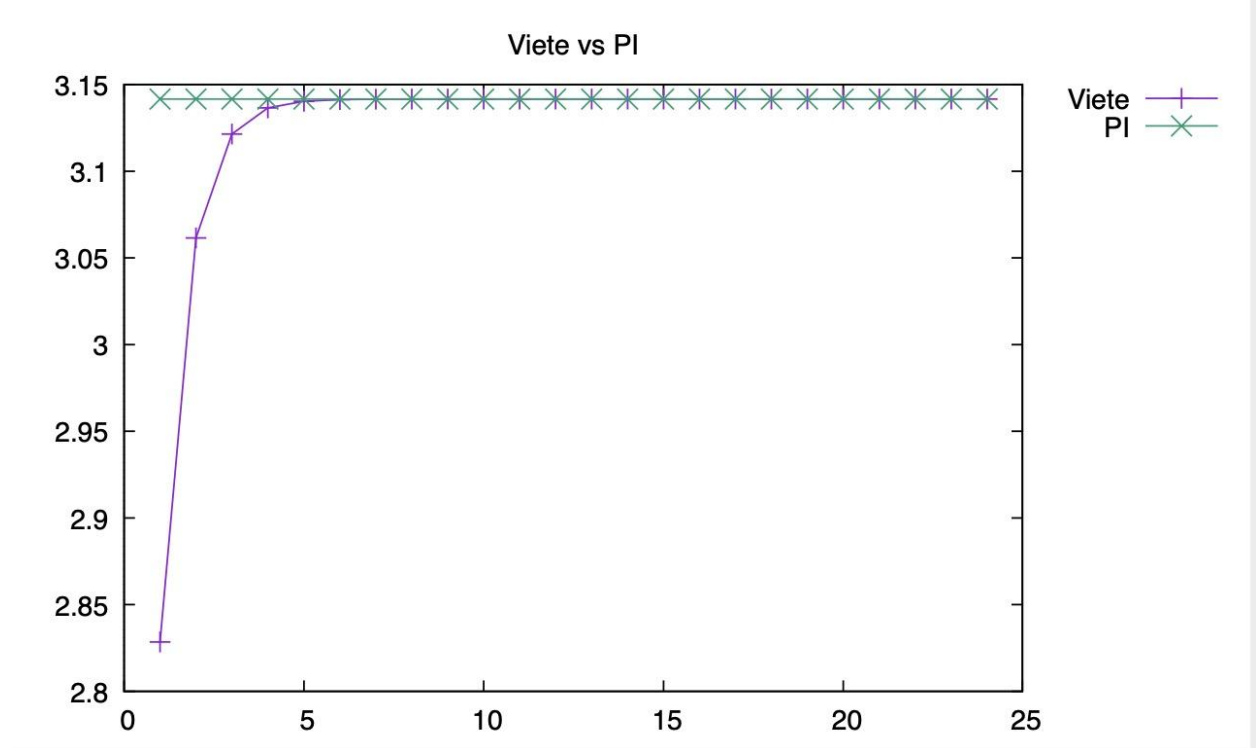
This program implements a small number of mathematical functions, which mimic those of `<math.h>`, and uses them to compute the fundamental constant e and π . In this writeup, I will present my results of comparing my functions to those of the math library's.

Results:

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Viete vs pi:

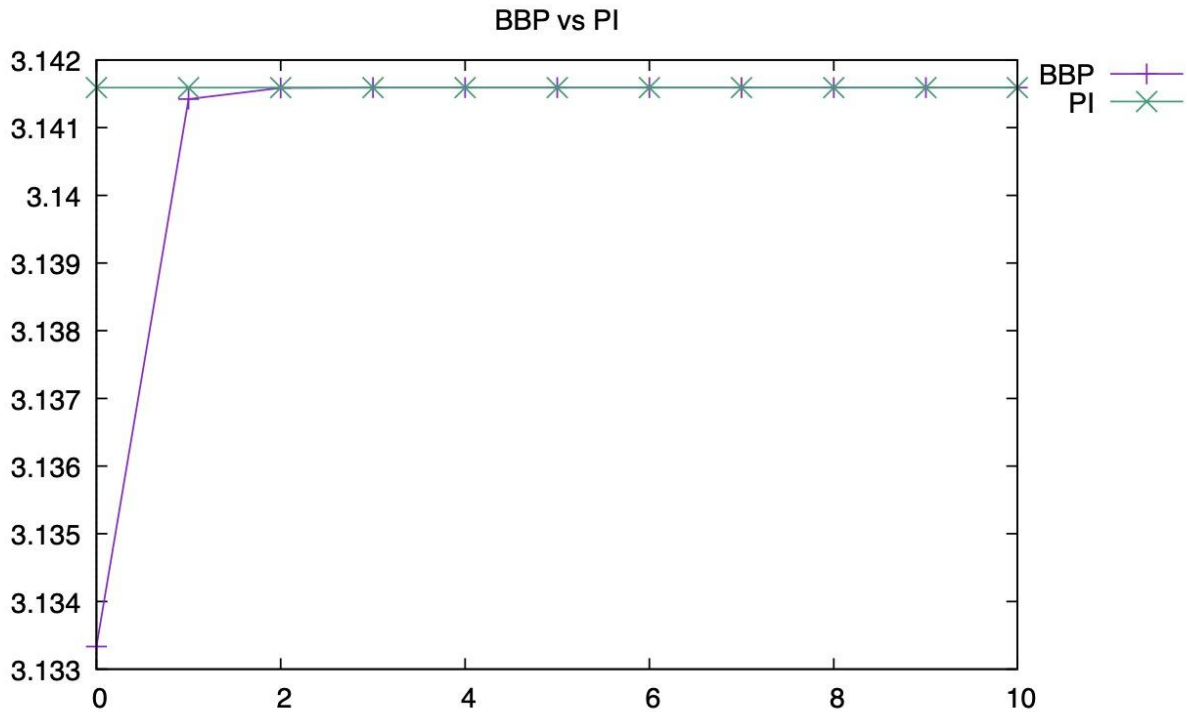
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The x axis represents the number of iterations of my function. The y axis represents the approximated value of pi. As you can see, the green line represents the math library's constant pi. My function takes 23 iterations to get to pi. The difference between constant pi and my viete function is 0.0000000000000004.

BBP vs pi:

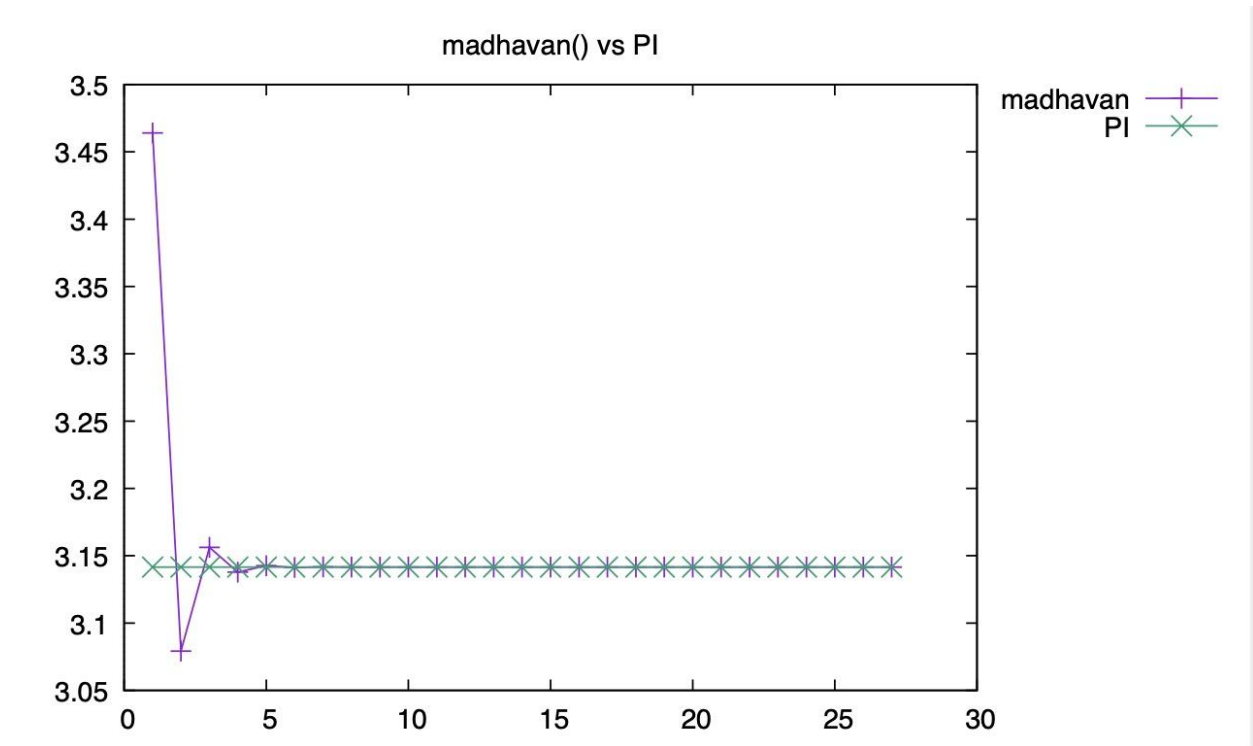
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The x axis represents the number of iterations of my function. The y axis represents the approximated value of pi. As you can see, the green line represents the math library's constant pi. My function takes 11 iterations to get to pi. There is no difference between constant pi and my bbp function.

Madhava vs pi:

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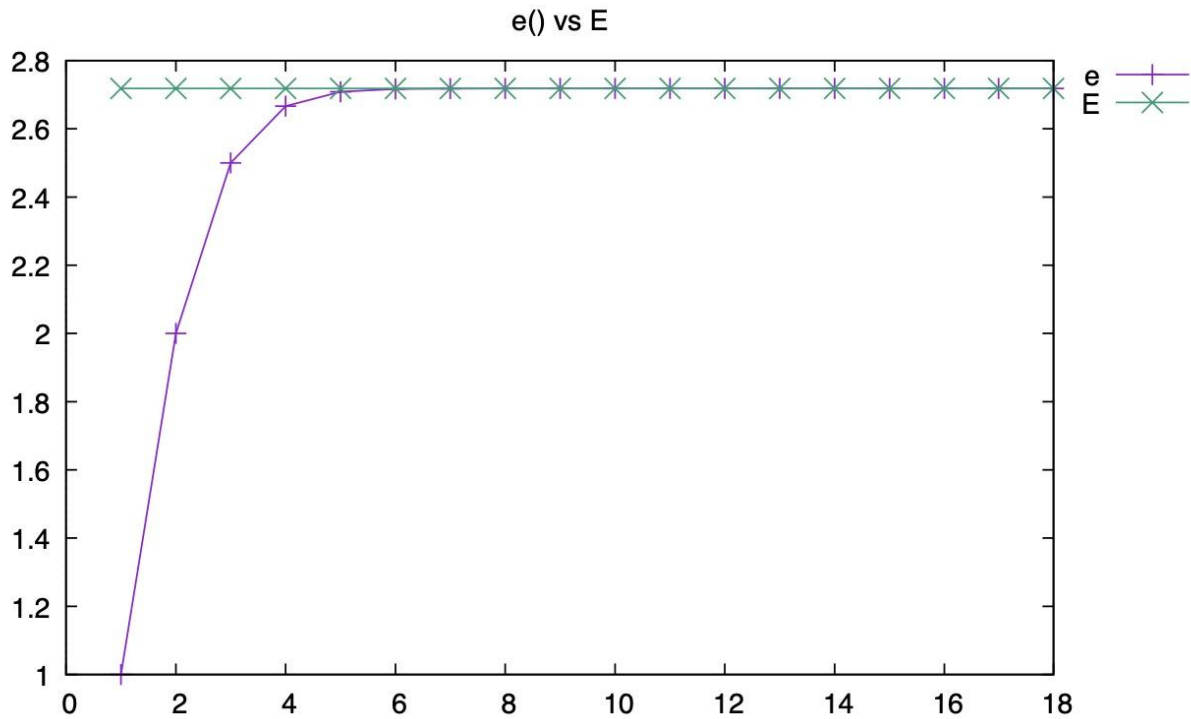


There is a typo on the graph. It should be comparing *madhava vs pi.

The x axis represents the number of iterations of my function. The y axis represents the approximated value of pi. As you can see, the green line represents the math library's constant pi. My function takes 27 iterations to get to pi. There is a difference of 0.0000000000000007 between constant pi and my bbp function.

e() vs constant M_E:

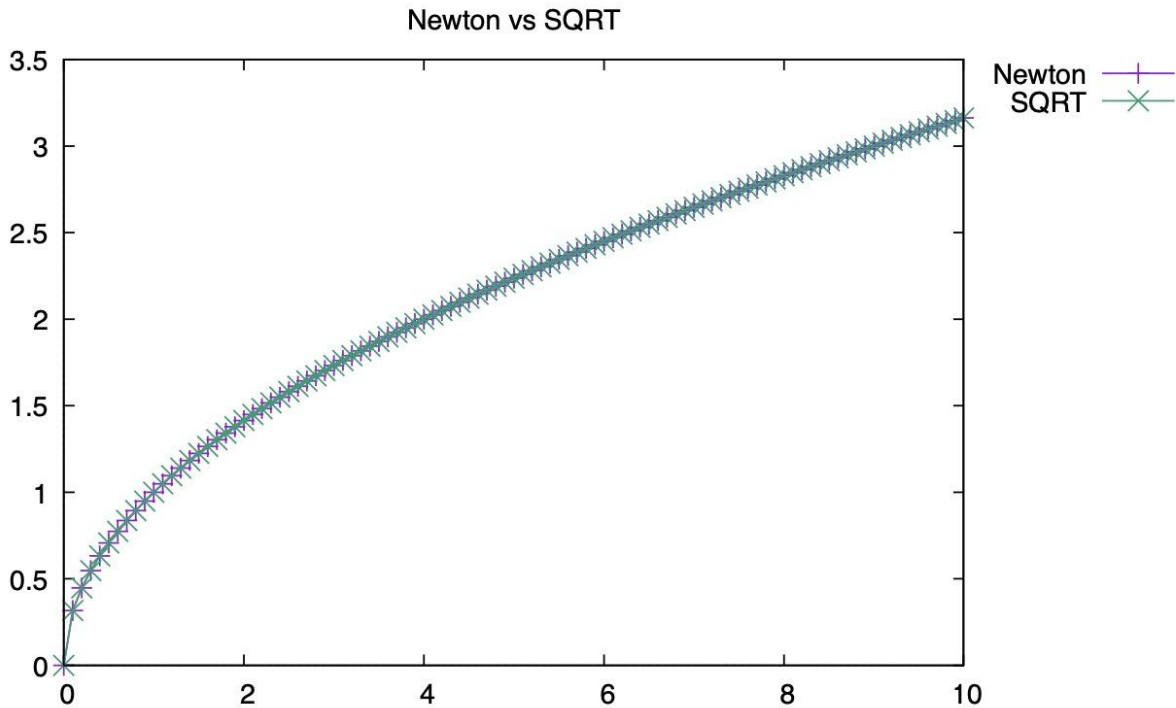
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The x axis represents the number of iterations of my function. The y axis represents the approximated value of e. As you can see, the green line represents the math library's constant E. My function takes 18 iterations to get to pi. There is no difference between constant pi and my e function.

Newton vs sqrt:

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This graph compares my `newton_sqrt` function vs math library's `sqrt` function. The x axis represents the number to be square rooted from 0.0 to 10.0, and the y axis represents the approximated value of that square root. For each number from 0.0 to 10.0, my approximation matched the math library's exactly.

Problems:

I ran into an issue with plotting the euler graph. There were 9948880 terms and my gnuplot crashed when I tried to input the data. As for my results, my approximation had a difference of 0.000000095981660 with the math library's.